Machine Learning Lab - Lab 2

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1 Understanding Bootstrap Evaluation with a Decision Tree Classifier

1.1 Definition of Bootstrapping

Bootstrapping is a statistical resampling technique used to estimate the accuracy or reliability of a model. Instead of splitting the dataset just once into training and testing sets, bootstrapping repeatedly samples from the data with replacement. This means that some data points may appear more than once in a sample, while others may not appear at all.

The data points that do not appear in a given bootstrap sample are called Out-of-Bag (OOB) samples. These OOB samples serve as a testing set, allowing us to evaluate the model's performance without needing a separate dataset. Over many iterations, we collect performance metrics and analyze their distribution. This gives us a more stable estimate of the model's true performance.

1.2 Procedure

The notebook implements bootstrapping by following these steps in each iteration:

1. Downloaded the dataset from `https://www.kaggle.com/datasets/uciml/iris`.
2. Load the dataset and check the first few entries to understand its structure.
3. Encode the categorical target variable into numerical format for compatibility with the decision tree classifier.
4. Create a bootstrap sample by randomly selecting data points with replacement.
5. The bootstrap sample is a random selection with replacement of data points from the original dataset. When there are repeatition in values, other values are left out, forming the OOB (Out-of-Bag) set.
6. The Decision Tree Classifier is trained on the bootstrap sample and tested on the OOB set.
7. The model makes predictions on the OOB samples and the accuracy of these predictions is calculated and stored.

8. This process is repeated for a specific number of iterations, each time generating a new bootstrap sample and evaluating the model on the corresponding OOB set.

9. After all iterations, the collected accuracy scores are analyzed to compute the mean accuracy and standard deviation.

10. A confidence interval for the accuracy scores is calculated to provide a range of expected performance.

11. Finally, a histogram of the accuracy scores is plotted to visualize their distribution across all

1.3 Evaluation Metrics

- **Mean Accuracy**: The average accuracy across all bootstrap iterations, providing an overall estimate of model performance.

- **Standard Deviation**: Measures the variability of accuracy scores, indicating how much the performance fluctuates across different bootstrap samples. A mathematical formula for standard deviation is:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$$

  where n is the number of samples, x_i are individual accuracy scores, and μ is the mean accuracy.

- **Confidence Interval**: A statistical range that likely contains the true accuracy of the model, giving insight into the reliability of the performance estimate.

- **Histogram of Accuracy Scores**: A visual representation showing the distribution of accuracy scores across all bootstrap iterations, helping to identify patterns or outliers in model performance.

1.4 Results and Interpretation

The following is a sample of the dataset used in the analysis:

Table 1: Sample of Iris Dataset

Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3.0	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5.0	3.6	1.4	0.2	Iris-setosa

Table 2: Species Distribution

Species	**Count**
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

After running the bootstrap evaluation with 1000 iterations, the following results were obtained:

- **Mean Accuracy**: 0.945
- **Standard Deviation**: 0.029
- **95% Confidence Interval**: (0.879, 1.0)

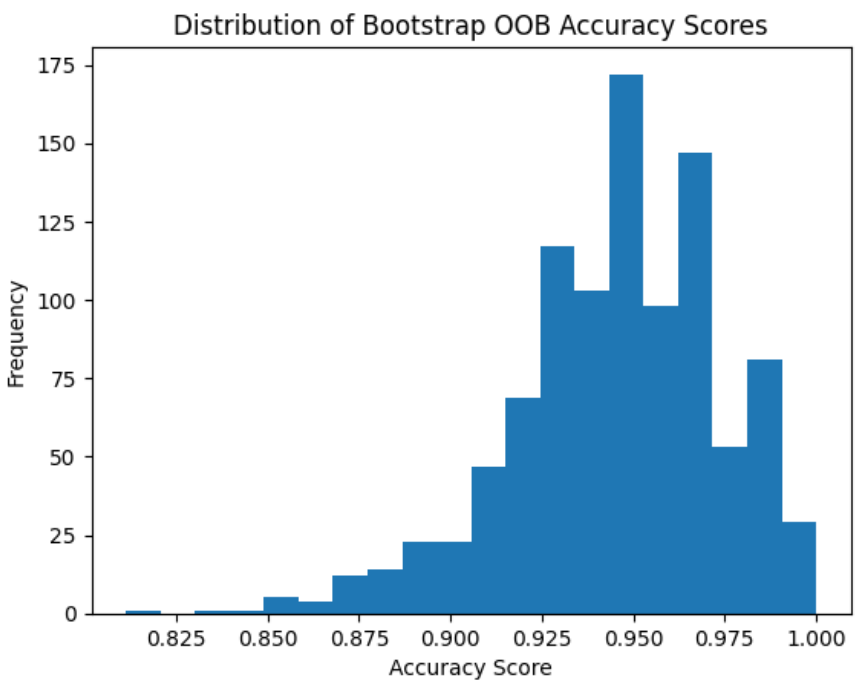

Figure 1: Histogram of Accuracy Scores from Bootstrap Evaluation

The histogram above shows the distribution of accuracy scores obtained from the bootstrap evaluation. The mean accuracy of 0.945 indicates that the decision tree classifier performs well on average across different bootstrap samples. The standard deviation of 0.029 suggests that there is some variability in performance, but it is relatively low, indicating consistent results.

The 95% confidence interval of (0.879, 1.0) suggests that we can be 95% confident that the true accuracy of the model lies within this range. This wide interval indicates some uncertainty in the estimate, likely due to the limited size of the dataset.

2 Understanding Different Interpretations using Percentage and Percentile Rankings

2.1 Definition of Percentage and Percentile

Percentage and percentile are two different statistical measures used to interpret data.

- **Percentage**: A percentage is a way of expressing a number as a fraction of 100. It is calculated by dividing the part by the whole and multiplying by 100. Percentages are commonly used to represent proportions, such as test scores or survey results.
- **Percentile**: A percentile is a measure that indicates the relative standing of a value within a dataset. It represents the percentage of data points that fall below a specific value. For example, if a student scores in the 90th percentile on a test, it means they scored higher than 90% of the other students who took the test.

2.1.1 Percentage Formula

The percentage is calculated using the following formula:

$$\text{Percentage} = \frac{\text{Obtained Marks}}{\text{Total Marks}} \times 100$$

For example, if a student obtains 85 marks out of 100, their percentage is:

$$\text{Percentage} = \frac{85}{100} \times 100 = 85\%$$

2.1.2 Percentile Formula

The percentile rank of a student is calculated using the following formula:

$$\text{Percentile} = \frac{\text{Number of students with marks below the student}}{\text{Total number of students}} \times 100$$

Alternatively, using the position method:

$$\text{Percentile Rank} = \frac{P}{N} \times 100$$

where P is the position of the student when arranged in ascending order, and N is the total number of students.

For example, if a student ranks 9th out of 10 students, their percentile rank is:

$$\text{Percentile Rank} = \frac{9}{10} \times 100 = 90\text{th percentile}$$

2.2 Implications of Using Percentage vs. Percentile

- load the dataset from marksheet.xls, the data is in csv format, so just change the file extinsion to .csv.
- Calculate the total marks for each student by summing their scores across all subjects.
- create functions to convert total marks to precentage and percentile, then to grade and label of each one.
- Train the Decision Tree Classifier for grade prediction, then another one for percentage label prediction, then another one for percentile label prediction.
- Evaluate and compare the performance of the models using confusion matrices and accuracy scores.
- Cross check the test of both percentage and percentile and plot the confusion matrices for both.

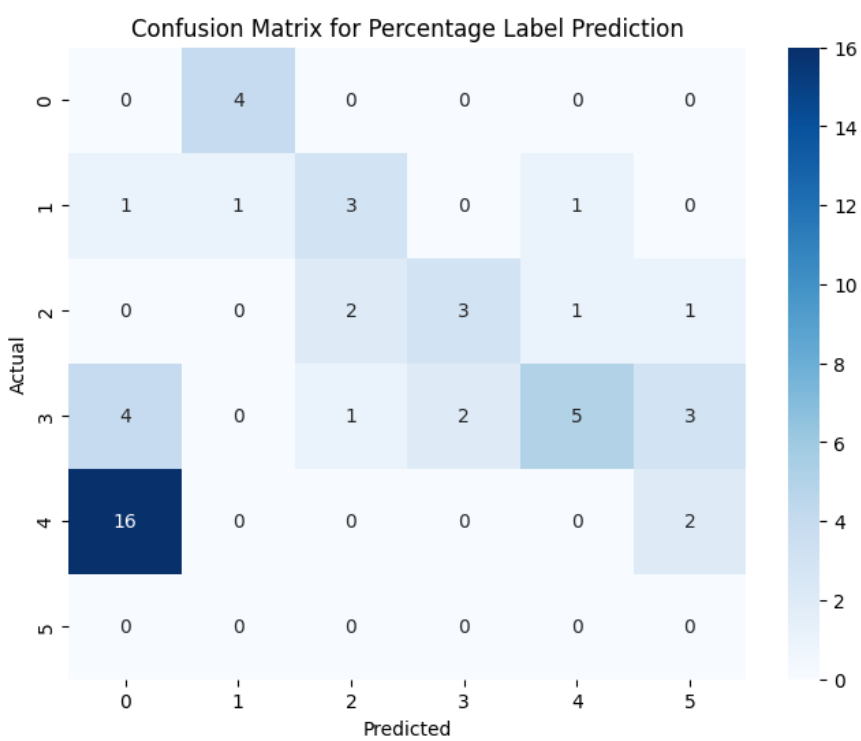

Figure 2: Confusion Matrix for Percentage Label Prediction

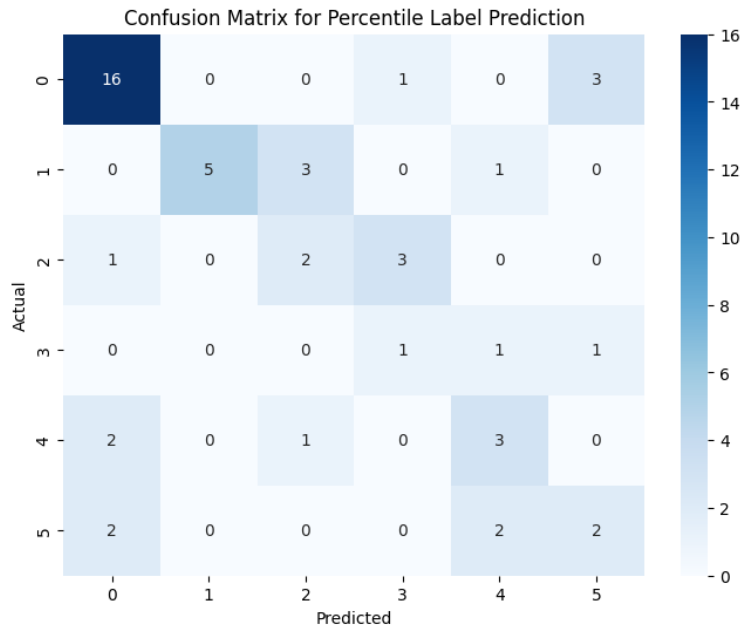

Figure 3: Confusion Matrix for Percentile Label Prediction

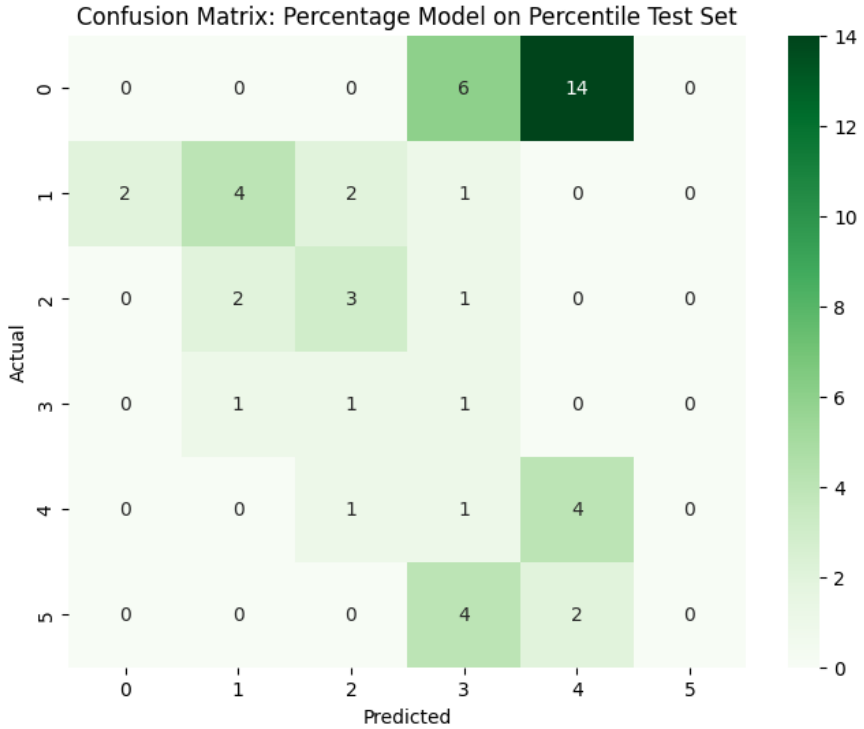

Figure 4: Confusion Matrix for Percentage Model on Percentile Test Data

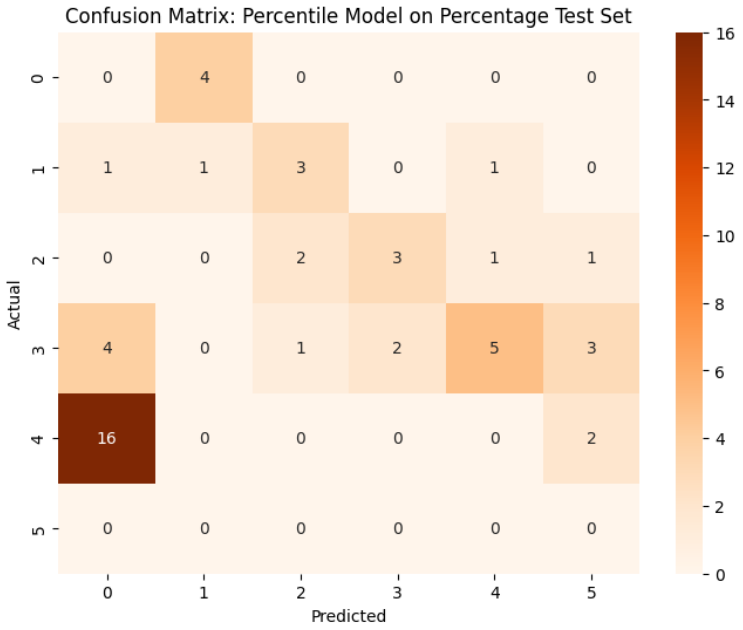

Figure 5: Confusion Matrix for Percentile Model on Percentage Test Data